1.а)

$$\mathbb{E}\xi = -1\cdot\tfrac{1}{4} + 0\cdot\tfrac{3}{8} + 1\cdot\tfrac{1}{8} + 2\cdot\tfrac{1}{4} = -\tfrac{2}{8} + \tfrac{1}{8} + \tfrac{4}{8} = \frac{3}{8}.$$

$$\mathbb{E}\xi^2 = 1\cdot\tfrac{1}{4} + 0\cdot\tfrac{3}{8} + 1\cdot\tfrac{1}{8} + 4\cdot\tfrac{1}{4} = \tfrac{2}{8} + \tfrac{1}{8} + \tfrac{8}{8} = \frac{11}{8}.$$

$$\mathbb{D}\xi = \frac{11}{8} - \frac{9}{64} = \frac{88-9}{64} = \boxed{\frac{79}{64}}, \qquad \boxed{\mathbb{E}\xi = \frac{3}{8}}.$$

1.б)

F_ξ ступенчатая $\Rightarrow$ ξ дискретна; значения — точки скачков, вероятности — величины скачков:

$$P(\xi = -1) = \tfrac{1}{2} - 0 = \tfrac{1}{2}, \qquad P(\xi = 2) = 1 - \tfrac{1}{2} = \tfrac{1}{2}.$$

$$\mathbb{E}\xi = -1\cdot\tfrac{1}{2} + 2\cdot\tfrac{1}{2} = \boxed{\frac{1}{2}}, \qquad \mathbb{E}\xi^2 = 1\cdot\tfrac{1}{2} + 4\cdot\tfrac{1}{2} = \frac{5}{2}.$$

$$\mathbb{D}\xi = \frac{5}{2} - \frac{1}{4} = \boxed{\frac{9}{4}}.$$

1.в)

(распределение Пуассона)

$$\mathbb{E}\xi = \sum_{k=0}^{\infty} k\,\frac{\lambda^k}{k!}e^{-\lambda} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda}e^{\lambda} = \boxed{\lambda}.$$

$\mathbb{E}\xi^2$ удобнее считать через $\mathbb{E}\xi(\xi-1)$:

$$\mathbb{E}\xi(\xi-1) = \sum_{k=2}^{\infty} k(k-1)\frac{\lambda^k}{k!}e^{-\lambda} = \lambda^2 e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!} = \lambda^2.$$

$$\mathbb{E}\xi^2 = \mathbb{E}\xi(\xi-1) + \mathbb{E}\xi = \lambda^2 + \lambda, \qquad \mathbb{D}\xi = \lambda^2 + \lambda - \lambda^2 = \boxed{\lambda}.$$

2.а)

Проверка нормировки:

$$\int_{-1}^{0} \tfrac{1}{2}\,dx + \int_{0}^{\infty} e^{-x/2}\,dx = \frac{1}{2} + 2 = \frac{5}{2} \neq 1.$$

Это не плотность распределения.

2.б)

F_ξ непрерывна в точке 2: $\dfrac{2^3}{C} = 1 \;\Rightarrow\; C = 8.$

$$p_\xi(x) = F_\xi'(x) = \frac{3x^2}{8}, \quad 0 < x \le 2.$$

$$\mathbb{E}\xi = \int_0^2 x\cdot\frac{3x^2}{8}\,dx = \frac{3}{8}\cdot\frac{x^4}{4}\Big|_0^2 = \frac{3}{8}\cdot 4 = \boxed{\frac{3}{2}}.$$

$$\mathbb{E}\xi^2 = \int_0^2 \frac{3x^4}{8}\,dx = \frac{3}{8}\cdot\frac{32}{5} = \frac{12}{5}, \qquad \mathbb{D}\xi = \frac{12}{5} - \frac{9}{4} = \frac{48-45}{20} = \boxed{\frac{3}{20}}.$$

2.в)

Нормировка:

$$\int_{-\pi}^{\pi} C|\sin x|\,dx = 2C\int_0^{\pi}\sin x\,dx = 2C\cdot 2 = 4C = 1 \;\Rightarrow\; C = \frac{1}{4}.$$

Плотность чётная $\Rightarrow x\,p_\xi(x)$ нечётна:

$$\mathbb{E}\xi = \int_{-\pi}^{\pi}\frac{x|\sin x|}{4}\,dx = \boxed{0}.$$

$$\mathbb{E}\xi^2 = 2\cdot\frac{1}{4}\int_0^{\pi} x^2\sin x\,dx.$$

По частям:

$$\int_0^{\pi} x^2\sin x\,dx = \big[-x^2\cos x\big]_0^{\pi}+2\int_0^{\pi} x\cos x\,dx = \pi^2+2\Big(\big[x\sin x\big]_0^{\pi}-\int_0^{\pi}\sin x\,dx\Big) = \pi^2+2(0-2) = \pi^2-4.$$

$$\mathbb{E}\xi^2 = \frac{\pi^2-4}{2}, \qquad \mathbb{D}\xi = \boxed{\frac{\pi^2-4}{2}}.$$

2.г)

$$p_\xi(x) = F'_\xi(x) = \cos x, \quad 0\le x\le\tfrac{\pi}{2}.$$

$$\mathbb{E}\xi = \int_0^{\pi/2} x\cos x\,dx = \big[x\sin x\big]_0^{\pi/2} - \int_0^{\pi/2}\sin x\,dx = \frac{\pi}{2}-1.$$

$$\mathbb{E}\xi^2 = \int_0^{\pi/2} x^2\cos x\,dx = \big[x^2\sin x\big]_0^{\pi/2}-2\int_0^{\pi/2} x\sin x\,dx = \frac{\pi^2}{4}-2\Big(\big[-x\cos x\big]_0^{\pi/2}+\int_0^{\pi/2}\cos x\,dx\Big) = \frac{\pi^2}{4}-2.$$

$$\mathbb{D}\xi = \frac{\pi^2}{4}-2-\Big(\frac{\pi}{2}-1\Big)^2 = \frac{\pi^2}{4}-2-\frac{\pi^2}{4}+\pi-1 = \boxed{\pi-3}, \qquad \boxed{\mathbb{E}\xi = \frac{\pi}{2}-1}.$$

2.д)

(гамма-распределение) Сразу считаем момент $\mathbb{E}\xi^k$, замена $t = ax$:

$$\mathbb{E}\xi^k = \frac{a^p}{\Gamma(p)}\int_0^{\infty} x^{p+k-1}e^{-ax}\,dx = \frac{a^p}{\Gamma(p)}\cdot\frac{\Gamma(p+k)}{a^{p+k}} = \frac{\Gamma(p+k)}{\Gamma(p)\,a^k}.$$

С учётом $\Gamma(p+1) = p\Gamma(p)$, $\Gamma(p+2) = (p+1)p\,\Gamma(p)$:

$$\mathbb{E}\xi = \frac{p\,\Gamma(p)}{\Gamma(p)\,a} = \boxed{\frac{p}{a}}, \qquad \mathbb{E}\xi^2 = \frac{p(p+1)}{a^2}.$$

$$\mathbb{D}\xi = \frac{p(p+1)}{a^2}-\frac{p^2}{a^2} = \boxed{\frac{p}{a^2}}.$$

2.е)

$$p_\xi(x) = F'_\xi(x) = \frac{1}{x^2}, \quad x>1.$$

$$\mathbb{E}\xi = \int_1^{\infty} x\cdot\frac{1}{x^2}\,dx = \int_1^{\infty}\frac{dx}{x} = \ln x\Big|_1^{\infty} = +\infty.$$

Интеграл расходится $\Rightarrow$ **математическое ожидание не существует**, а значит не существует и дисперсия.

3.а)

$\eta = \xi^2$, $\xi \sim U[-1,1]$.

$$\mathbb{E}\eta = \mathbb{E}\xi^2 = \int_{-1}^{1} \frac{x^2}{2}\,dx = \frac{x^3}{6}\Big|_{-1}^{1} = \boxed{\frac{1}{3}}.$$

$$\mathbb{E}\eta^2 = \mathbb{E}\xi^4 = \int_{-1}^{1} \frac{x^4}{2}\,dx = \frac{x^5}{10}\Big|_{-1}^{1} = \frac{1}{5}, \qquad \mathbb{D}\eta = \frac{1}{5} - \frac{1}{9} = \boxed{\frac{4}{45}}.$$

3.б)

$\eta = e^{\xi - 1}$.

Замена $t = x - a$:

$$\mathbb{E}\eta = \int_a^{\infty} e^{x-1}\,\lambda e^{-\lambda(x-a)}\,dx = \lambda e^{a-1}\int_0^{\infty} e^{-(\lambda-1)t}\,dt = \boxed{\frac{\lambda\, e^{a-1}}{\lambda - 1}}, \qquad \lambda > 1.$$

При $\lambda \le 1$ интеграл расходится: $\mathbb{E}\eta = +\infty$.

$$\mathbb{E}\eta^2 = \int_a^{\infty} e^{2(x-1)}\,\lambda e^{-\lambda(x-a)}\,dx = \lambda e^{2(a-1)}\int_0^{\infty} e^{-(\lambda-2)t}\,dt = \frac{\lambda\, e^{2(a-1)}}{\lambda - 2}, \qquad \lambda > 2.$$

$$\mathbb{D}\eta = \lambda e^{2(a-1)}\left(\frac{1}{\lambda-2} - \frac{\lambda}{(\lambda-1)^2}\right) = \lambda e^{2(a-1)}\,\frac{(\lambda-1)^2 - \lambda(\lambda-2)}{(\lambda-2)(\lambda-1)^2} = \boxed{\frac{\lambda\, e^{2(a-1)}}{(\lambda-2)(\lambda-1)^2}}, \qquad \lambda > 2.$$

При $1 < \lambda \le 2$: $\mathbb{E}\eta$ конечно, но $\mathbb{D}\eta = +\infty$.